

performance analogue-to-digital converters and commonly-used fixed-function peripherals. The PSoC family is made up of 8-bit (PSoC1 and PSoC3) and 32-bit MCUs (PSoC4 and PSoC5). These have flash memories up to 256kB, SRAM up to 64kB and internal EEPROM up to 2kB. (Flash can be used to emulate EEPROM.) PSoC works in ultra-low power mode. In this mode, it consumes less than 1µA current, which is useful for operating in standby mode.

Using PSoC Creator IDE tool, all interface and logic can be designed. This tool is a readily-available component block for designing interface and logic like SARADC and PGA for analogue sensors and other inputs. PWM, CLK, MUX and comparator components are used for motor-drive applications. Character LCD and segment LCD components directly (do not require external buffers) drive LCD/graphical LCDs.

The tool also has a real-time clock component for real-time measurement. It has an internal system clock and so does not require external clock/oscillator circuitry. Its other components include a timer, buzzer, infrared and fan controller.

PSoC Creator IDE tool enables you to make use of an entire tools ecosystem with integrated compiler tool chains, RTOS solutions and production programmers. With this tool, you can create and share user-defined custom peripherals using hierarchical schematic designs. You can automatically place and route chosen components and integrate simple glue logic, normally located in discrete devices.

A PSoC supports CapSense technology, which replaces mechanical buttons with a CapSense based keypad. This reduces failure due to mechanical buttons and provides better product reliability. PSoC Creator IDE tool supports CapSense

SmartSense component, which auto-tunes the sensitivity of CapSense buttons and slider (does not require manual tuning).

A PSoC supports the waterproofing application. It also supports proximity function for the front keypad; the front keypad gets activated when you place your hand near the keypad.

A PSoC can directly drive LED and LCD displays (segment LCDs, graphic LCDs and character LCD displays) for displaying channel and volume information. It has an operating range of 1.71V to 5.5V, so it can be easily interfaced with external peripherals for other applications.

A PSoC works in ultra-low power mode. In this mode it consumes less than 1µA current, which is useful for low-power-battery based applications.

A PSoC has internal PGA, comparators and configurable delta-sigma ADC with 8- to 20-bit resolution,



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